Technical Appendix

This appendix documents how the dataset was assembled and how the models were estimated. It reproduces the main analytical workflow from:

- `data_merging.ipynb` for dataset construction and crosswalk logic
- `models/causal_analysis.ipynb` for sample creation, weighted descriptive analysis, parallel-trends checks, DiD estimation, and robustness checks

Appendix A. Data Assembly Code

Below shows how the CPS data were merged with industry-level AI exposure (`AIIE`) and occupation-level AI exposure (`AIOE`). Because CPS ASEC uses three different Census industry vintages across 2016–2025 (2012 codes for 2016–2019, 2017 codes for 2020–2024, 2022 codes for 2025), the assembly builds a single unified AIIE lookup that covers all three vintages via alias rows so every year resolves against the same table.

In [21]:
```
import os
from pathlib import Path
import sys
import pandas as pd
import warnings

warnings.filterwarnings("ignore")
os.environ["PYTHONWARNINGS"] = "ignore"

for candidate in [Path.cwd(), Path.cwd().parent]:
    if (candidate / "notebook_helpers.py").exists():
        if str(candidate) not in sys.path:
            sys.path.insert(0, str(candidate))
        REPO_ROOT = candidate
        break
else:
    raise FileNotFoundError("Could not find notebook_helpers.py")

from notebook_helpers import (
    ind_to_aiie,
    occ_to_aioe,
    parse_naics_patterns,
    parse_soc_pattern,
    to_canonical_ind,
)

aiie = pd.read_excel("../data/AIOE_DataAppendix.xlsx", sheet_name="Appendix
B")

crosswalk_clean = (
    pd.read_excel(
        "../data/2022-Census-Industry-Code-List-with-Crosswalk.xlsx",
        sheet_name="2022 Census Ind Code List ",
    )
    .dropna(how="all")
    .rename(
        columns={
            "Industry 2022 Description": "industry_desc",
            "2022 Census Industry Code": "census_ind_raw",
            "2022 NAICS Code": "naics_raw",
        }
    )
)

is_leaf = ~crosswalk_clean["census_ind_raw"].astype(str).str.contains("-",
na=False)
```

```
crosswalk_clean = crosswalk_clean.loc[is_leaf].copy()
crosswalk_clean["IND"] = pd.to_numeric(
    crosswalk_clean["census_ind_raw"], errors="coerce"
).astype("Int64")
crosswalk_clean = crosswalk_clean.dropna(subset=["IND"])

crosswalk_clean["naics_patterns"] = crosswalk_clean["naics_raw"].apply(
    parse_naics_patterns
)

# 2017 -> 2022: ffill both sides so split- and merge-continuation rows
aren't dropped,
# then dedupe (blank separator rows between groups otherwise leak
duplicates).
cc_17 = pd.read_excel(
    "../data/2022-Census-Industry-Code-List-with-Crosswalk.xlsx",
    sheet_name="2022 Ind Code Changes",
    header=2,
)
cc_17[["2017 Census Industry Code", "2017 Census Industry Description"]] =
cc_17[
    ["2017 Census Industry Code", "2017 Census Industry Description"]
].ffill()
cc_17[["2022 Census Industry Code", "2022 Census Industry Description"]] =
cc_17[
    ["2022 Census Industry Code", "2022 Census Industry Description"]
].ffill()
cc_17 = cc_17.dropna(
    subset=["2017 Census Industry Code", "2022 Census Industry Code"]
).astype({"2017 Census Industry Code": int, "2022 Census Industry Code":
int})
cc_17 = cc_17.drop_duplicates(
    subset=["2017 Census Industry Code", "2022 Census Industry Code"]
)

n_children_17 = cc_17.groupby("2017 Census Industry Code").size()
singletons_17 = n_children_17[n_children_17 == 1].index
splits_17 = n_children_17[n_children_17 > 1].index
ind_2017_to_2022 = dict(
    zip(
        cc_17.loc[
            cc_17["2017 Census Industry Code"].isin(singletons_17),
            "2017 Census Industry Code",
        ],
        cc_17.loc[
            cc_17["2017 Census Industry Code"].isin(singletons_17),
            "2022 Census Industry Code",
        ],
    )
)
ind_2017_splits = {
    parent: grp["2022 Census Industry Code"].tolist()
    for parent, grp in cc_17[
        cc_17["2017 Census Industry Code"].isin(splits_17)
    ].groupby("2017 Census Industry Code")
}

# 2012 -> 2017: CPS ASEC 2016-2019 data uses 2012 Census codes. This was an
error
cc_12 = pd.read_excel(
    "../data/2017-industry-code-list-with-crosswalk.xlsx",
    sheet_name="Summary of Changes",
    header=2,
).rename(columns=lambda c: c.strip())
cc_12[["2012 Census Industry Code", "2012 Description"]] = cc_12[
    ["2012 Census Industry Code", "2012 Description"]
].ffill()
cc_12[["2017 Census Industry Code", "2017 Description"]] = cc_12[
    ["2017 Census Industry Code", "2017 Description"]
].ffill()
cc_12 = cc_12.dropna(
    subset=["2012 Census Industry Code", "2017 Census Industry Code"]
).astype({"2012 Census Industry Code": int, "2017 Census Industry Code":
```

```
int})
cc_12 = cc_12.drop_duplicates(
    subset=["2012 Census Industry Code", "2017 Census Industry Code"]
)

n_children_12 = cc_12.groupby("2012 Census Industry Code").size()
singletons_12 = n_children_12[n_children_12 == 1].index
splits_12 = n_children_12[n_children_12 > 1].index
ind_2012_to_2017 = dict(
    zip(
        cc_12.loc[
            cc_12["2012 Census Industry Code"].isin(singletons_12),
            "2012 Census Industry Code",
        ],
        cc_12.loc[
            cc_12["2012 Census Industry Code"].isin(singletons_12),
            "2017 Census Industry Code",
        ],
    )
)
ind_2012_splits = {
    parent: grp["2017 Census Industry Code"].tolist()
    for parent, grp in cc_12[
        cc_12["2012 Census Industry Code"].isin(splits_12)
    ].groupby("2012 Census Industry Code")
}
```

In [22]:

```
aiie_lookup = dict(zip(aiie["NAICS"].astype(str), aiie["AIIE"]))
aiie_keys_longest_first = sorted(aiie_lookup, key=len, reverse=True)

crosswalk_clean["AIIE"] = crosswalk_clean["naics_patterns"].apply(
    lambda patterns: ind_to_aiie(patterns, aiie_lookup,
aiie_keys_longest_first)
)
ind_aiie = (
    crosswalk_clean.dropna(subset=["AIIE"])
    .astype({"IND": "int64"})[["IND", "industry_desc", "AIIE"]]
    .reset_index(drop=True)
)

# Add alias rows so 2017- and 2012-vintage CPS codes resolve against the
same table.
# 1:1 renames/merges inherit the child's AIIE; split parents get the mean
of children.
aiie_by_2022 = dict(zip(ind_aiie["IND"], ind_aiie["AIIE"]))

alias_2017 = []
for c17, c22 in ind_2017_to_2022.items():
    if c17 in aiie_by_2022:
        continue
    if c22 in aiie_by_2022:
        alias_2017.append(
            {"IND": c17, "industry_desc": "(2017 alias)", "AIIE":
aiie_by_2022[c22]}
        )
for parent, kids in ind_2017_splits.items():
    vals = [aiie_by_2022[k] for k in kids if k in aiie_by_2022]
    if vals:
        alias_2017.append(
            {
                "IND": parent,
                "industry_desc": f"(2017 parent of {len(kids)} 2022
codes)",
                "AIIE": sum(vals) / len(vals),
            }
        )

aiie_by_2017 = dict(aiie_by_2022)
for row in alias_2017:
    aiie_by_2017[row["IND"]] = row["AIIE"]

alias_2012 = []
for c12, c17 in ind_2012_to_2017.items():
```

```
    if c12 in aiie_by_2017:
        continue
    if c17 in aiie_by_2017:
        alias_2012.append(
            {"IND": c12, "industry_desc": "(2012 alias)", "AIIE":
aiie_by_2017[c17]}
        )
for parent, kids in ind_2012_splits.items():
    vals = [aiie_by_2017[k] for k in kids if k in aiie_by_2017]
    if vals:
        alias_2012.append(
            {
                "IND": parent,
                "industry_desc": f"(2012 parent of {len(kids)} 2017
codes)",
                "AIIE": sum(vals) / len(vals),
            }
        )

ind_aiie = pd.concat(
    [ind_aiie, pd.DataFrame(alias_2017), pd.DataFrame(alias_2012)],
    ignore_index=True,
)

df = pd.read_csv("../data/cps_00001.csv")

df["IND_2022"] = df["IND"].map(
    lambda x: to_canonical_ind(x, ind_2012_to_2017, ind_2017_to_2022)
)
df_aiie = df.merge(ind_aiie, on="IND", how="left")
```

In [23]:

```
df_aiie.head()
```

Out[23]:

	YEAR	SERIAL	MONTH	CPSID	ASECFLAG	ASECWTH	PERNUM	
0	2016	2	3	20141202933700	1	1,132.0500	1	201412
1	2016	7	3	20151202935900	1	1,077.5400	1	201512
2	2016	7	3	20151202935900	1	1,077.5400	2	201512
3	2016	8	3	20160202937400	1	1,151.4900	1	201602
4	2016	9	3	20160102910900	1	705.0000	1	20160

In [24]:

```
aioe = pd.read_excel("../data/AIOE_DataAppendix.xlsx", sheet_name="Appendix
A")

occ_crosswalk = (
    pd.read_excel(
        "../data/2018-occupation-code-list-and-crosswalk.xlsx",
        sheet_name="2018 Census Occ Code List",
        header=0,
    )
    .dropna(how="all")
    .rename(
        columns={
            "2018 Census Title ": "occ_desc",
            "2018 Census Code": "census_occ_raw",
            "2018 SOC Code": "soc_raw",
        }
    )
)

is_leaf = ~occ_crosswalk["census_occ_raw"].astype(str).str.contains("-",
na=False)
occ_crosswalk = occ_crosswalk.loc[is_leaf].copy()
occ_crosswalk["OCC"] = pd.to_numeric(
    occ_crosswalk["census_occ_raw"], errors="coerce"
).astype("Int64")
occ_crosswalk = occ_crosswalk.dropna(subset=["OCC"])
```

```
occ_crosswalk["soc_pattern"] =
occ_crosswalk["soc_raw"].apply(parse_soc_pattern)

aioe_lookup = dict(zip(aioe["SOC Code"].astype(str), aioe["AIOE"]))
aioe_keys_longest_first = sorted(aioe_lookup, key=len, reverse=True)

occ_crosswalk["AIOE"] = occ_crosswalk["soc_pattern"].apply(
    lambda pattern: occ_to_aioe(pattern, aioe_lookup,
aioe_keys_longest_first)
)
occ_aioe = (
    occ_crosswalk.dropna(subset=["AIOE"])
    .astype({"OCC": "int64"})[["OCC", "occ_desc", "AIOE"]]
    .reset_index(drop=True)
)
```

In [25]:
```
occ_aioe.head()
```

Out[25]:

	OCC	occ_desc	AIOE
0	10	Chief executives	1.3342
1	20	General and operations managers	0.5749
2	40	Advertising and promotions managers	1.2944
3	51	Marketing managers	1.3150
4	52	Sales managers	1.2663

In [26]:
```
occ_changes = (
    pd.read_excel(
        "../data/2018-occupation-code-list-and-crosswalk.xlsx",
        sheet_name="Occ Code Changes",
        header=2,
    )
    .rename(
        columns={
            "2010 Census OCC Code": "occ_2010",
            "2010 Census Title": "title_2010",
            "New 2018 Census OCC Code": "occ_2018",
        }
    )
    .dropna(how="all")
)

occ_changes[["occ_2010", "title_2010"]] = occ_changes[
    ["occ_2010", "title_2010"]
].ffill()
occ_changes = occ_changes.dropna(subset=["occ_2010", "occ_2018"]).astype(
    {"occ_2010": int, "occ_2018": int}
)

occ_2018_to_aioe = dict(zip(occ_aioe["OCC"], occ_aioe["AIOE"]))
occ_changes["AIOE"] = occ_changes["occ_2018"].map(occ_2018_to_aioe)

pre2018 = (
    occ_changes[~occ_changes["occ_2010"].isin(occ_aioe["OCC"])]
    .dropna(subset=["AIOE"])
    .groupby("occ_2010", as_index=False)
    .agg(occ_desc=("title_2010", "first"), AIOE=("AIOE", "mean"))
    .rename(columns={"occ_2010": "OCC"})
)

occ_aioe_full = pd.concat([occ_aioe, pre2018], ignore_index=True)
df_aiie = df_aiie.merge(occ_aioe_full, on="OCC", how="left")

df_aiie.to_csv("../data/data_with_aiie.csv", index=False)
```

In [27]:
```
df_aiie.head()
```

Out[27]:

	YEAR	SERIAL	MONTH	CPSID	ASECFLAG	ASECWTH	PERNUM	
0	2016	2	3	20141202933700	1	1,132.0500	1	201412
1	2016	7	3	20151202935900	1	1,077.5400	1	201512
2	2016	7	3	20151202935900	1	1,077.5400	2	201512
3	2016	8	3	20160202937400	1	1,151.4900	1	201602
4	2016	9	3	20160102910900	1	705.0000	1	20160

Appendix B. Analysis Setup and Variable Construction

The next blocks reproduce the final causal analysis setup, including reproducibility settings, sample restrictions, variable definitions, and helper functions used throughout the analysis.

Analysis Frame and Key Variables

The final analysis uses CPS ASEC repeated cross-sections from 2016-2025 merged with industry-level AI exposure, `AIIE`. The analytical frame applies the following restrictions and variable definitions:

- Working-age sample: ages 25-64
- Exposure requirement: non-missing `AIIE`
- Unemployment sample: individuals in the labor force
- Wage sample: employed individuals with positive wage income
- Main wage DiD sample: same as the wage sample, excluding 2023 as a transition year
- Survey weights: `ASECWT`
- Clustered standard errors: by `IND_2022`
- `post_unemployment = 1` for `YEAR >= 2023`
- `post_wage = 1` for `YEAR >= 2024`
- `early_career = 1` for ages 25-34
- `log_wage = log(INCWAGE)` for positive wage observations

For plots only, `AIIE` is also split into low- and high-exposure groups using the 25th and 75th percentiles. The regressions use continuous `AIIE`, not the grouped version.

In [28]:

```
from pathlib import Path
import random
import sys

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

for candidate in [Path.cwd(), Path.cwd().parent]:
    if (candidate / "notebook_helpers.py").exists():
        if str(candidate) not in sys.path:
            sys.path.insert(0, str(candidate))
        REPO_ROOT = candidate
        break
else:
    raise FileNotFoundError("Could not find notebook_helpers.py")

from notebook_helpers import (
    coef_table,
    extract_event_study_table,
    fit_event_study,
    joint_zero_test,
    plot_event_study,
    plot_event_study_on_axis,
```

```
    plot_trends,
    run_wls,
    summarize_sample,
    wald_difference_test,
    weighted_mean,
    weighted_mean_se,
    yearly_group_stats,
)

SEED = 42
random.seed(SEED)
np.random.seed(SEED)

plt.style.use("seaborn-v0_8-whitegrid")
pd.set_option("display.max_columns", 100)
pd.set_option("display.float_format", lambda x: f"{x:,.4f}")

DATA_PATH = Path("../data/data_with_aiie.csv")
data = pd.read_csv(DATA_PATH)
```

In [29]:

```
df = data.copy()

df = df[df["AGE"].between(25, 64)].copy()
df = df[df["AIIE"].notna()].copy()

df["employed"] = df["EMPSTAT"].isin([10, 12]).astype(int)
df["unemployed"] = df["EMPSTAT"].isin([20, 21, 22]).astype(int)
df["in_labor_force"] = (df["LABFORCE"] == 2).astype(int)

df["post_unemployment"] = (df["YEAR"] >= 2023).astype(int)
df["post_wage"] = (df["YEAR"] >= 2024).astype(int)

df["early_career"] = df["AGE"].between(25, 34).astype(int)
df["older_worker"] = df["AGE"].between(50, 64).astype(int)

df["positive_wage"] = (df["INCWAGE"] > 0).astype(int)
df["log_wage"] = np.where(df["INCWAGE"] > 0, np.log(df["INCWAGE"]), np.nan)

q1 = df["AIIE"].quantile(0.25)
q3 = df["AIIE"].quantile(0.75)
df["exposure_group"] = np.select(
    [df["AIIE"] <= q1, df["AIIE"] >= q3],
    ["Low exposure", "High exposure"],
    default="Middle exposure",
)

unemp_sample = df[df["in_labor_force"] == 1].copy()
wage_sample = df[(df["employed"] == 1) & (df["positive_wage"] == 1)].copy()
wage_did_sample = wage_sample[wage_sample["YEAR"] != 2023].copy()
```

In [30]:

```
from notebook_helpers import (
    weighted_mean,
    weighted_mean_se,
    summarize_sample,
    yearly_group_stats,
    smooth_segment,
    plot_trends,
    run_wls,
    coef_table,
    wald_difference_test,
)
```

Appendix C. Core Estimation Code

This section reproduces the main estimation workflow from the causal analysis notebook. It starts with the model specifications, then shows the sample checks, descriptive plots, event-study diagnostics, main difference-in-differences regressions, and age-heterogeneity specifications.

Difference-in-Differences Specifications

The baseline unemployment difference-in-differences model is:

$$\begin{aligned} unemployed_{it} = \beta_0 &+ \beta_1 AIIE_j + \beta_2 post_t^{unemp} \\ &+ \beta_3 \left(AIIE_j \times post_t^{unemp}\right) + \beta_4 AGE_i \\ &+ \sum_e \delta_e 1\{EDUC_i = e\} + \varepsilon_{it} \end{aligned}$$

The unemployment heterogeneity model adds a triple interaction:

$$\begin{aligned} unemployed_{it} = \alpha_0 &+ \alpha_1 AIIE_j + \alpha_2 post_t^{unemp} + \alpha_3 early_career_i \\ &+ \alpha_4 \left(AIIE_j \times post_t^{unemp}\right) \\ &+ \alpha_5 \left(AIIE_j \times early_career_i\right) \\ &+ \alpha_6 \left(post_t^{unemp} \times early_career_i\right) \\ &+ \alpha_7 \left(AIIE_j \times post_t^{unemp} \times early_career_i\right) \\ &+ \alpha_8 AGE_i + \sum_e \rho_e 1\{EDUC_i = e\} + u_{it} \end{aligned}$$

The baseline wage difference-in-differences model is:

$$\begin{aligned} \log(wage_{it}) = \gamma_0 &+ \gamma_1 AIIE_j + \gamma_2 post_t^{wage} \\ &+ \gamma_3 \left(AIIE_j \times post_t^{wage}\right) + \gamma_4 AGE_i \\ &+ \sum_e \kappa_e 1\{EDUC_i = e\} + \nu_{it} \end{aligned}$$

The wage heterogeneity model parallels the unemployment triple-interaction setup:

$$\begin{aligned} \log(wage_{it}) = \eta_0 &+ \eta_1 AIIE_j + \eta_2 post_t^{wage} + \eta_3 early_career_i \\ &+ \eta_4 \left(AIIE_j \times post_t^{wage}\right) \\ &+ \eta_5 \left(AIIE_j \times early_career_i\right) \\ &+ \eta_6 \left(post_t^{wage} \times early_career_i\right) \\ &+ \eta_7 \left(AIIE_j \times post_t^{wage} \times early_career_i\right) \\ &+ \eta_8 AGE_i + \sum_e \phi_e 1\{EDUC_i = e\} + \omega_{it} \end{aligned}$$

Variable definitions:

- $unemployed_{it}$: indicator equal to 1 if individual i is unemployed in year t
- $\log(wage_{it})$: log wage income for individual i in year t
- $AIIE_j$: AI Industry Exposure for industry j
- $post_t^{unemp}$: indicator equal to 1 in 2023-2025
- $post_t^{wage}$: indicator equal to 1 in 2024-2025
- $early_career_i$: indicator equal to 1 for ages 25-34
- AGE_i: age in years
- $1\{EDUC_i = e\}$: education-category controls

All models are estimated using CPS ASEC weights and cluster-robust standard errors at the 2022 Census industry level.

Sample Checks and Descriptive Diagnostics

These checks reproduce the summary tables used to verify the analysis samples before estimating the descriptive plots and regressions.

In [31]:
```
display(summarize_sample(unemp_sample, ["AIIE", "unemployed", "AGE"]))
display(summarize_sample(wage_sample, ["AIIE", "INCWAGE", "log_wage",
"AGE"]))

display(unemp_sample.groupby("YEAR").size().rename("unemployment_sample_n")
.to_frame())
```

```
display(wage_did_sample.groupby("YEAR").size().rename("wage_sample_n").to_frame())
```

	variable	mean	std	min	max	missing
0	AIIE	0.1365	1.1567	-1.9986	2.2156	0
1	unemployed	0.0362	0.1867	0.0000	1.0000	0
2	AGE	43.1917	10.7862	25.0000	64.0000	0

	variable	mean	std	min	max	missing
0	AIIE	0.1940	1.1496	-1.9986	2.2156	0
1	INCWAGE	67,534.2531	81,493.3720	2.0000	2,099,999.0000	0
2	log_wage	10.7564	0.8956	0.6931	14.5574	0
3	AGE	43.0997	10.7663	25.0000	64.0000	0

YEAR	unemployment_sample_n
2016	63686
2017	64460
2018	62313
2019	63020
2020	54754
2022	52850
2023	50876
2024	50831
2025	50052

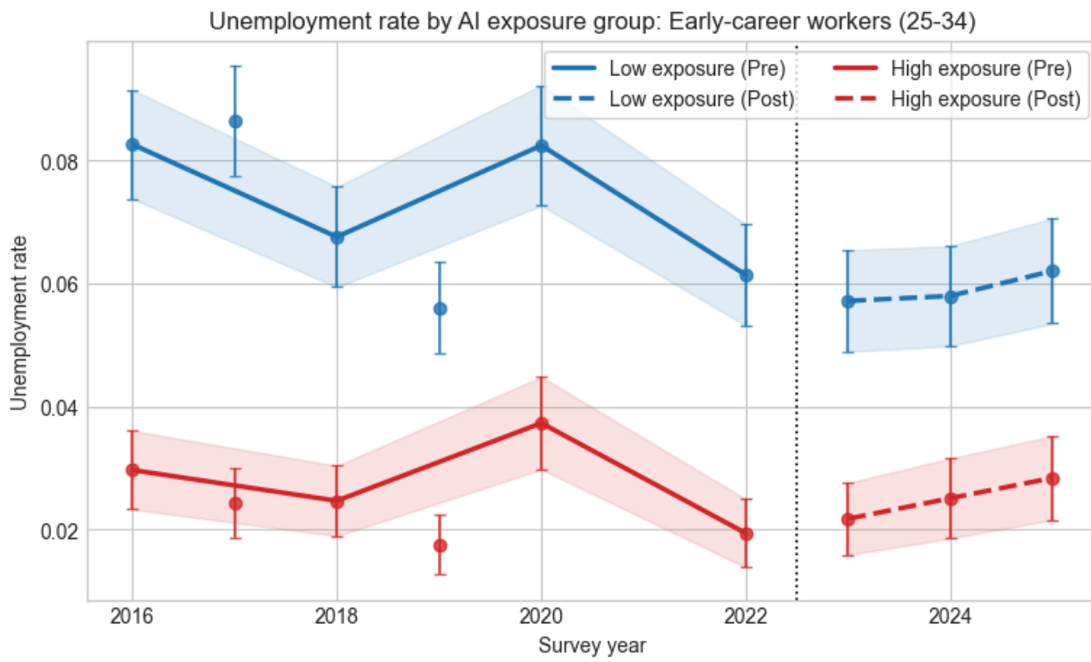
YEAR	wage_sample_n
2016	55908
2017	56881
2018	55256
2019	56276
2020	48316
2022	47023
2024	45466
2025	44847

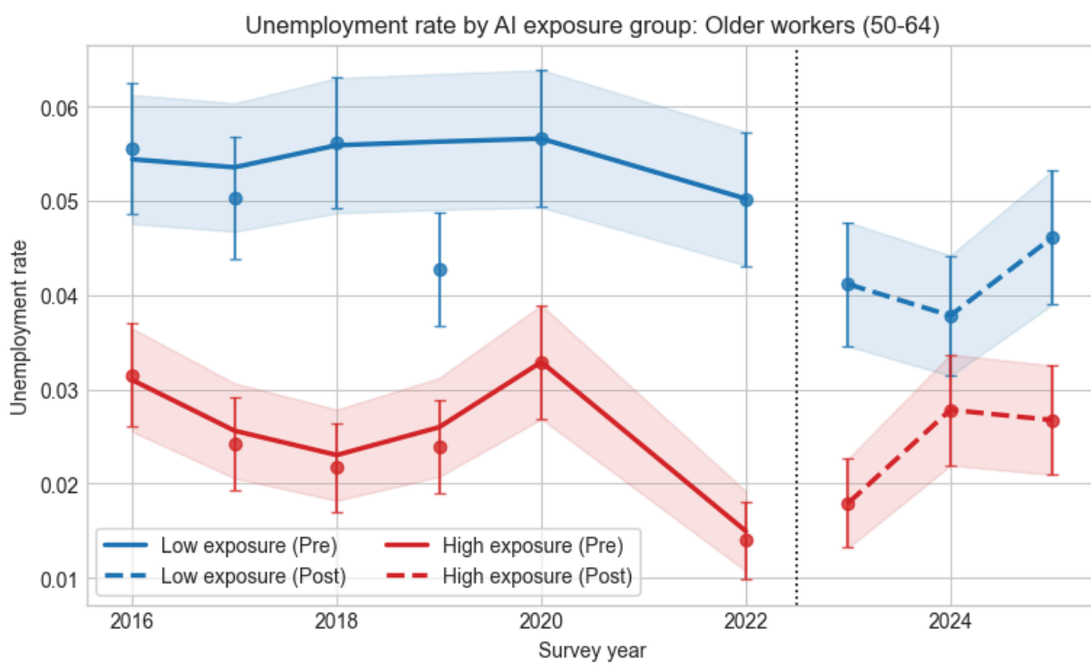
Age-Specific Descriptive Plots

The analysis notebook also checks whether the broad unemployment pattern looks similar when focusing on early-career workers and on an older comparison group.

In [32]:
```
figure_specs = [
    (
        "Early-career workers (25-34)",
        unemp_sample[unemp_sample["early_career"] == 1],
    ),
    (
        "Older workers (50-64)",
        unemp_sample[unemp_sample["older_worker"] == 1],
    ),
```

```
]

for label, subset in figure_specs:
    fig, ax, _ = plot_trends(
        subset,
        value_col="unemployed",
        title=f"Unemployment rate by AI exposure group: {label}",
        ylabel="Unemployment rate",
        post_start=2023,
    )
    plt.show()
```

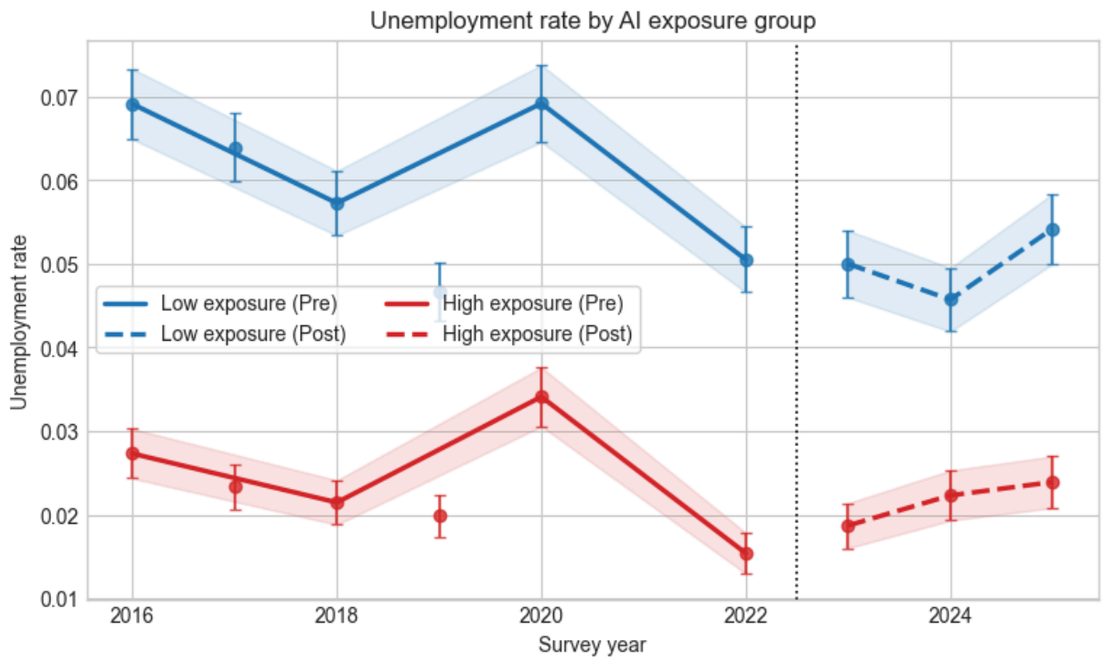


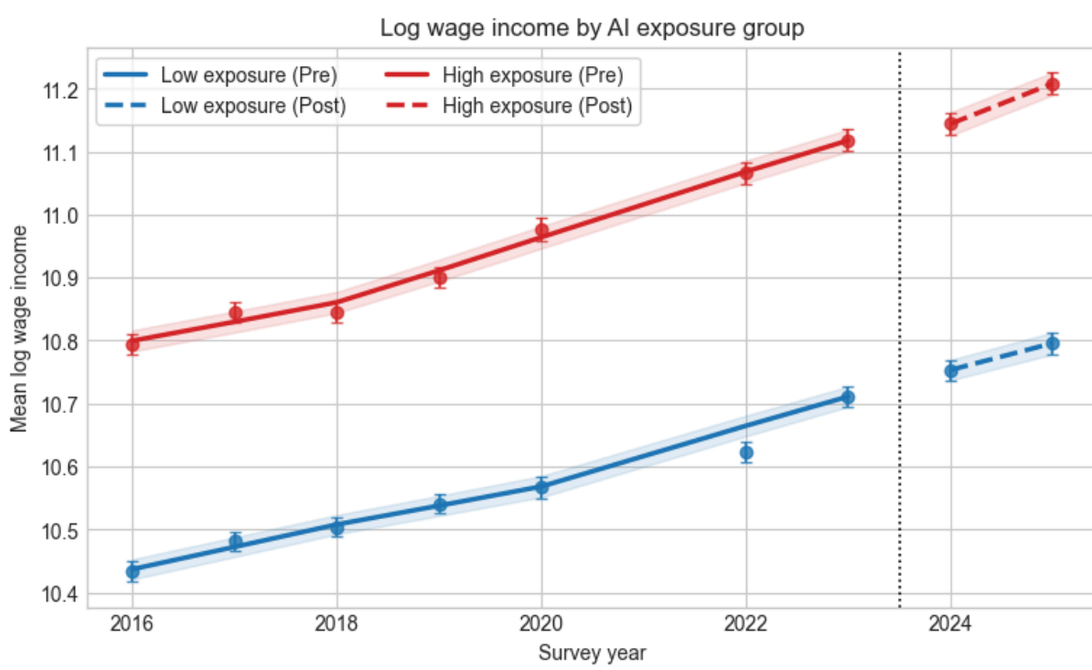


In [33]:

```
coverage_table = pd.DataFrame(
    {
        "sample": [
            "Working-age analysis frame",
            "Unemployment sample",
            "Wage sample",
            "Main wage DiD sample",
        ],
        "n_rows": [len(df), len(unemp_sample), len(wage_sample),
len(wage_did_sample)],
    }
)
coverage_table

fig_unemp, ax_unemp, unemp_plot_df = plot_trends(
    unemp_sample,
    value_col="unemployed",
```

```
    title="Unemployment rate by AI exposure group",
    ylabel="Unemployment rate",
    post_start=2023,
)
plt.show()

fig_wage, ax_wage, wage_plot_df = plot_trends(
    wage_sample,
    value_col="log_wage",
    title="Log wage income by AI exposure group",
    ylabel="Mean log wage income",
    post_start=2024,
)
plt.show()
```





Why Check Parallel Trends?

The descriptive figures are useful for showing the broad pattern in the data, but they are not enough on their own to justify a difference-in-differences interpretation. The key identifying assumption behind the DiD design is that, absent the post-2022 expansion of generative AI, outcomes in higher- and lower-exposure industries would have continued to move along similar underlying paths.

This is why we now dive into an event-study version of the model. The event-study check asks whether the relationship between `AIIE` and the outcome was already shifting before the post period began. If those pre-period differences are small, that

makes the DiD setup more credible. If they are large or systematic, then the post-period DiD estimates are harder to interpret causally because some of the gap may reflect pre-existing divergence rather than a post-2022 change.

Parallel Trends and Event-Study Diagnostic

Parallel trends are evaluated with an event-study specification that replaces the single post-period interaction with year-specific interactions between `AIIE` and survey year:

$$
\begin{aligned}
Y_{it} = \theta_0 &+ \sum_{\tau \neq 2022} \theta_\tau \left(AIIE_j \times 1\{YEAR_t = \tau\} \right) \\
&+ \gamma_j + \sum_{\tau \neq 2022} \lambda_\tau 1\{YEAR_t = \tau\} \\
&+ \sum_k \psi_k X_{kit} + \varepsilon_{it}
\end{aligned}
$$

where Y_{it} is either unemployment or log wage income, and X_{kit} includes age and education controls. This is not a staggered-adoption design. The omitted year is 2022, so each coefficient shows how the relationship between `AIIE` and the outcome differs relative to 2022.

Interpretation of the diagnostic:

- If pre-period coefficients are close to zero, that supports the parallel-trends assumption.
- If pre-period coefficients are systematically different from zero, that suggests pre-existing differential trends.
- The joint pre-trend test evaluates whether all pre-period `year × AIIE` terms are zero together.

Neither outcome fully passes this diagnostic. The unemployment specification still looks more informative than the wage specification because its pre-period deviations are smaller and less systematic, but it should still be treated as suggestive rather than definitive.

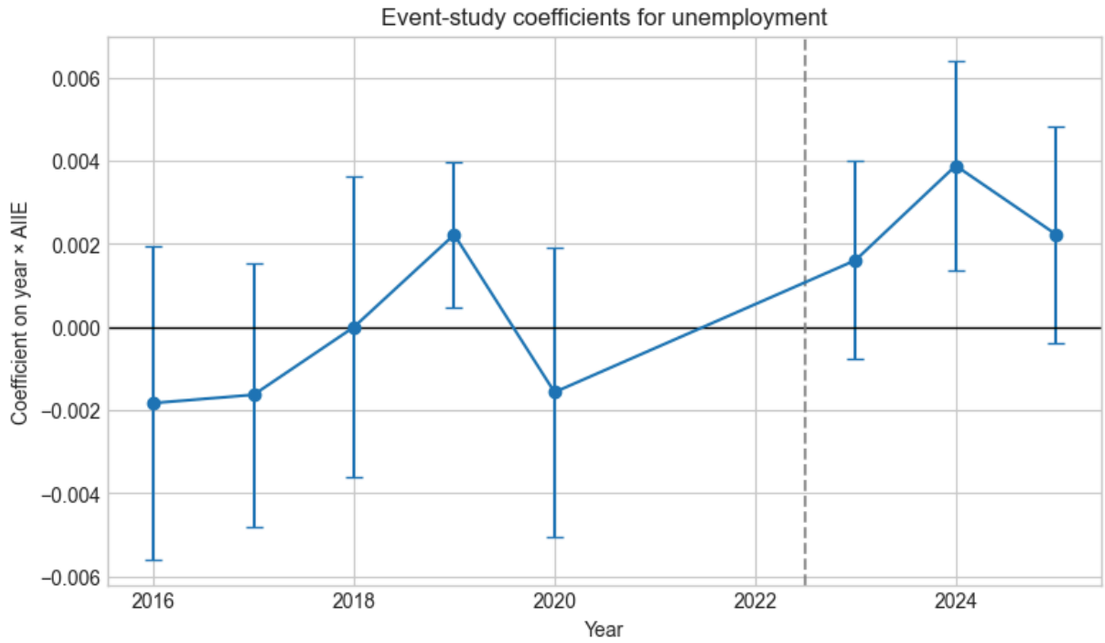
The event-study coefficient tables already report point estimates with `95%` confidence intervals and p-values. The joint pre-trend tests remain summarized with p-values because they test several coefficients at once, rather than a single parameter with one confidence interval.

In [34]:
```
from notebook_helpers import (
    fit_event_study,
    extract_event_study_table,
    joint_zero_test,
    plot_event_study_on_axis,
    plot_event_study,
)
```

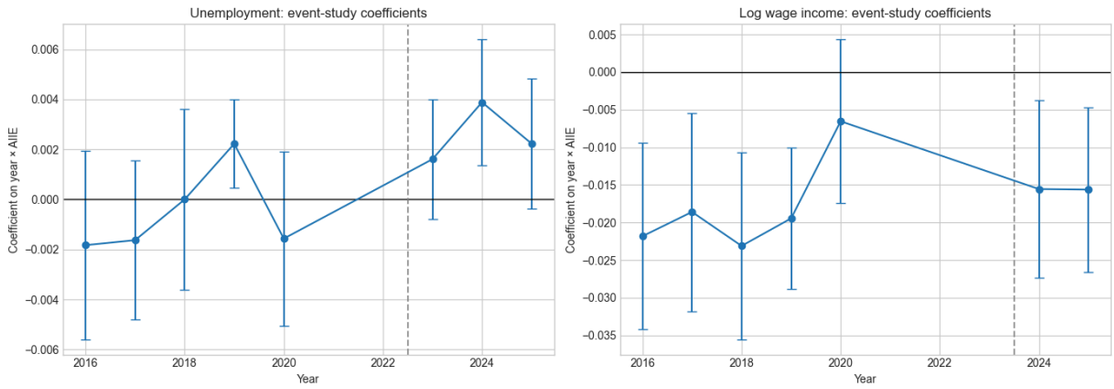
In [35]:
```
unemp_event = fit_event_study(unemp_sample, "unemployed",
reference_year=2022)
unemp_event_table = extract_event_study_table(unemp_event,
reference_year=2022)
display(unemp_event_table)

fig_unemp_event, ax_unemp_event = plot_event_study(
    unemp_event_table,
    title="Event-study coefficients for unemployment",
    post_start=2023,
)
plt.show()

unemp_pre_terms = unemp_event_table.loc[
    unemp_event_table["year"] < 2023, "term"
].tolist()
```

```
joint_zero_test(unemp_event, unemp_pre_terms)

wage_event = fit_event_study(wage_did_sample, "log_wage",
reference_year=2022)
wage_event_table = extract_event_study_table(wage_event,
reference_year=2022)
display(wage_event_table)

fig_wage_event, ax_wage_event = plot_event_study(
    wage_event_table,
    title="Event-study coefficients for log wage income",
    post_start=2024,
)
plt.show()

wage_pre_terms = wage_event_table.loc[
    wage_event_table["year"] < 2024, "term"
].tolist()
joint_zero_test(wage_event, wage_pre_terms)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
plot_event_study_on_axis(
    unemp_event_table,
    ax1,
    "Unemployment: event-study coefficients",
    post_start=2023,
)
plot_event_study_on_axis(
    wage_event_table,
    ax2,
    "Log wage income: event-study coefficients",
    post_start=2024,
)
plt.tight_layout()
plt.show()
```

	term	year	coef	std_err	ci_low	ci_high	p_value
0	C(YEAR, Treatment(reference=2022))[T.2016]:AIIE	2016	-0.0018	0.0019	-0.0056	0.0019	0.3423
1	C(YEAR, Treatment(reference=2022))[T.2017]:AIIE	2017	-0.0016	0.0016	-0.0048	0.0015	0.3144
2	C(YEAR, Treatment(reference=2022))[T.2018]:AIIE	2018	0.0000	0.0018	-0.0036	0.0036	0.9996
3	C(YEAR, Treatment(reference=2022))[T.2019]:AIIE	2019	0.0022	0.0009	0.0005	0.0040	0.0131
4	C(YEAR, Treatment(reference=2022))[T.2020]:AIIE	2020	-0.0016	0.0018	-0.0051	0.0019	0.3792
5	C(YEAR, Treatment(reference=2022))[T.2023]:AIIE	2023	0.0016	0.0012	-0.0008	0.0040	0.1868
6	C(YEAR, Treatment(reference=2022))[T.2024]:AIIE	2024	0.0039	0.0013	0.0014	0.0064	0.0025
7	C(YEAR, Treatment(reference=2022))[T.2025]:AIIE	2025	0.0022	0.0013	-0.0004	0.0048	0.0934



	term	year	coef	std_err	ci_low	ci_high	p_value
0	C(YEAR, Treatment(reference=2022))[T.2016]:AIIE	2016	-0.0218	0.0063	-0.0342	-0.0094	0.0006
1	C(YEAR, Treatment(reference=2022))[T.2017]:AIIE	2017	-0.0186	0.0067	-0.0318	-0.0054	0.0057
2	C(YEAR, Treatment(reference=2022))[T.2018]:AIIE	2018	-0.0231	0.0063	-0.0355	-0.0107	0.0003
3	C(YEAR, Treatment(reference=2022))[T.2019]:AIIE	2019	-0.0194	0.0048	-0.0289	-0.0100	0.0001
4	C(YEAR, Treatment(reference=2022))[T.2020]:AIIE	2020	-0.0065	0.0056	-0.0174	0.0044	0.2401
5	C(YEAR, Treatment(reference=2022))[T.2024]:AIIE	2024	-0.0155	0.0060	-0.0274	-0.0037	0.0099
6	C(YEAR, Treatment(reference=2022))[T.2025]:AIIE	2025	-0.0156	0.0056	-0.0266	-0.0047	0.0052



In [36]:
```
print("Unemployment base DiD")
m_unemp_base = run_wls(
    "unemployed ~ AIIE * post_unemployment + AGE + C(EDUC)",
    unemp_sample,
)
display(coef_table(m_unemp_base, keep="AIIE|post"))

print("Unemployment heterogeneity DiD")
m_unemp_triple = run_wls(
    "unemployed ~ AIIE * post_unemployment * early_career + AGE + C(EDUC)",
    unemp_sample,
)
display(coef_table(m_unemp_triple, keep="AIIE|post|early"))

print("Wage base DiD")
m_wage_base = run_wls(
    "log_wage ~ AIIE * post_wage + AGE + C(EDUC)",
    wage_did_sample,
)
display(coef_table(m_wage_base, keep="AIIE|post"))

print("Wage heterogeneity DiD")
m_wage_triple = run_wls(
    "log_wage ~ AIIE * post_wage * early_career + AGE + C(EDUC)",
    wage_did_sample,
)
display(coef_table(m_wage_triple, keep="AIIE|post|early"))
```

```
Unemployment base DiD
```

	coef	std_err	p_value
AIIE	-0.0078	0.0015	0.0000
post_unemployment	-0.0046	0.0014	0.0015
AIIE:post_unemployment	0.0029	0.0012	0.0143

```
Unemployment heterogeneity DiD
```

	coef	std_err	p_value
AIIE	-0.0060	0.0017	0.0005
post_unemployment	-0.0040	0.0015	0.0080
AIIE:post_unemployment	0.0020	0.0013	0.1306
early_career	0.0083	0.0013	0.0000
AIIE:early_career	-0.0066	0.0013	0.0000
post_unemployment:early_career	-0.0019	0.0016	0.2318
AIIE:post_unemployment:early_career	0.0034	0.0013	0.0099

```
Wage base DiD
```

	coef	std_err	p_value
AIIE	0.0162	0.0459	0.7246
post_wage	0.2394	0.0072	0.0000
AIIE:post_wage	0.0026	0.0074	0.7297

```
Wage heterogeneity DiD
```

	coef	std_err	p_value
AIIE	0.0190	0.0469	0.6853
post_wage	0.2253	0.0075	0.0000
AIIE:post_wage	-0.0010	0.0081	0.8975
early_career	-0.1838	0.0126	0.0000
AIIE:early_career	-0.0087	0.0193	0.6512
post_wage:early_career	0.0484	0.0096	0.0000
AIIE:post_wage:early_career	0.0123	0.0077	0.1092

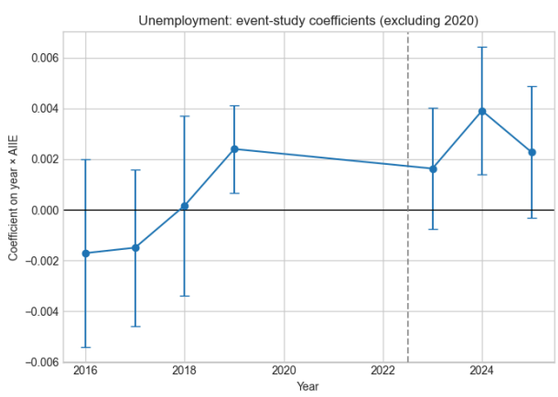
Appendix D. Robustness and Heterogeneity Code

The final blocks reproduce two follow-up checks from the analysis notebook: a robustness exercise that excludes 2020 and an age-split specification that asks whether the unemployment effect is more concentrated among workers ages 25-29 than among workers ages 30-34.

In [37]:
```
unemp_no20 = unemp_sample[unemp_sample["YEAR"] != 2020].copy()
wage_no20 = wage_did_sample[wage_did_sample["YEAR"] != 2020].copy()

m_unemp_no20 = run_wls(
    "unemployed ~ AIIE * post_unemployment * early_career + AGE + C(EDUC)",
    unemp_no20,
)
m_wage_no20 = run_wls(
    "log_wage ~ AIIE * post_wage * early_career + AGE + C(EDUC)",
    wage_no20,
)

display(coef_table(m_unemp_no20, keep="AIIE|post|early"))
display(coef_table(m_wage_no20, keep="AIIE|post|early"))

unemp_no20_event = fit_event_study(unemp_no20, "unemployed",
reference_year=2022)
unemp_no20_event_table = extract_event_study_table(unemp_no20_event,
reference_year=2022)
wage_no20_event = fit_event_study(wage_no20, "log_wage",
reference_year=2022)
wage_no20_event_table = extract_event_study_table(wage_no20_event,
reference_year=2022)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
plot_event_study_on_axis(
    unemp_no20_event_table,
    ax1,
    "Unemployment: event-study coefficients (excluding 2020)",
    post_start=2023,
)
plot_event_study_on_axis(
    wage_no20_event_table,
    ax2,
    "Log wage income: event-study coefficients (excluding 2020)",
    post_start=2024,
)
plt.tight_layout()
plt.show()
```

	coef	std_err	p_value
AIIE	-0.0058	0.0019	0.0019
post_unemployment	-0.0022	0.0016	0.1646
AIIE:post_unemployment	0.0016	0.0013	0.2140
early_career	0.0087	0.0015	0.0000
AIIE:early_career	-0.0069	0.0015	0.0000
post_unemployment:early_career	-0.0017	0.0017	0.2919
AIIE:post_unemployment:early_career	0.0037	0.0014	0.0078

	coef	std_err	p_value
AIIE	0.0167	0.0465	0.7194
post_wage	0.2346	0.0077	0.0000
AIIE:post_wage	0.0012	0.0085	0.8927
early_career	-0.1817	0.0137	0.0000
AIIE:early_career	-0.0085	0.0194	0.6615
post_wage:early_career	0.0469	0.0100	0.0000
AIIE:post_wage:early_career	0.0120	0.0083	0.1488



In [38]:
```
unemp_sample["age_group"] = pd.cut(
    unemp_sample["AGE"],
    bins=[24, 29, 34, 64],
    labels=["25-29", "30-34", "35-64"],
)

m_age_split = run_wls(
    "unemployed ~ AIIE * post_unemployment * C(age_group,
Treatment(reference='35-64'))"
    " + AGE + C(EDUC)",
    unemp_sample,
)

key_terms = [
    k
    for k in m_age_split.params.index
    if "AIIE:post_unemployment:C(age_group" in k
]
age_split_table = coef_table(m_age_split).loc[key_terms].copy()

term_25_29 = next(k for k in key_terms if "[T.25-29]" in k)
term_30_34 = next(k for k in key_terms if "[T.30-34]" in k)
age_gap_test = wald_difference_test(
    m_age_split,
    term_25_29,
    term_30_34,
    "25-29 coefficient minus 30-34 coefficient",
)
```

```
display(age_split_table)
display(age_gap_test)
```

	coef	std_err	p_value
AIIE:post_unemployment:C(age_group, Treatment(reference='35-64'))[T.25-29]	0.0045	0.0020	0.0274
AIIE:post_unemployment:C(age_group, Treatment(reference='35-64'))[T.30-34]	0.0024	0.0017	0.1712

	comparison	difference	wald_stat	p_value
0	25-29 coefficient minus 30-34 coefficient	0.0022	0.6477	0.4210

Interpreting the Age-Heterogeneity Test

This specification separates the younger portion of the sample into workers ages `25–29` and `30–34`, using workers ages `35–64` as the reference group. The interaction coefficients in `age_split_table` show whether the post-2022 unemployment relationship with `AIIE` is larger for each younger subgroup than for the reference group.

The `age_gap_test` output is the key comparison between the two younger subgroups. A positive difference means the `25–29` estimate is larger than the `30–34` estimate. In these results, the point estimate for ages `25–29` is larger, but the formal Wald test indicates that the two subgroup effects are not statistically different from one another. The most careful interpretation is that the unemployment effect appears somewhat stronger for the youngest workers, but the analysis does not provide strong enough evidence to claim that the `25–29` group is affected differently from the `30–34` group in a statistically distinct way.

Appendix E. Submission Export

The cell below uses `nbconvert` to export the notebook to HTML for submission.

In [39]:
```
from pathlib import Path
import subprocess
import sys

notebook = Path("appendix_data_science_memo.ipynb")
output_name = "appendix_data_science_memo"

cmd = [
    sys.executable,
    "-m",
    "jupyter",
    "nbconvert",
    "--to",
    "html",
    "--output",
    output_name,
    "--TagRemovePreprocessor.enabled=True",
    "--TagRemovePreprocessor.remove_cell_tags=remove_cell",
    "--log-level=ERROR",
    str(notebook),
]

print("Running:", " ".join(cmd))
subprocess.run(
    cmd, check=True, stdout=subprocess.PIPE, stderr=subprocess.PIPE,
text=True
)
print(f"Created {output_name}.html")
```

```
Running: /Users/aeshagandhi/Downloads/MIDS-Sp26/UDS/UDS-
Project/.venv/bin/python -m jupyter nbconvert --to html --output
appendix_data_science_memo --TagRemovePreprocessor.enabled=True --
TagRemovePreprocessor.remove_cell_tags=remove_cell --log-level=ERROR
appendix_data_science_memo.ipynb
Created appendix_data_science_memo.html
```

In [40]:
```
# # version without code cells

# notebook = Path("data_science_memo.ipynb")
# output_name = "no_code_data_science_memo_submission"

# cmd = [
#     sys.executable,
#     "-m",
#     "jupyter",
#     "nbconvert",
#     "--to",
#     "html",
#     "--output",
#     output_name,
#     "--no-input",
#     "--TagRemovePreprocessor.enabled=True",
#     "--TagRemovePreprocessor.remove_cell_tags=remove_cell",
#     "--log-level=ERROR",
#     str(notebook),
# ]

# print("Running:", " ".join(cmd))
# subprocess.run(
#     cmd, check=True, stdout=subprocess.PIPE, stderr=subprocess.PIPE,
text=True
# )
# print(f"Created {output_name}.html")
```

Appendix F. Sources

The project draws on the following primary data and reference sources:

Autor, David, Caroline Chin, Anna M. Salomons, and Bryan Seegmiller. *AI, Occupational Exposure, and the Future of Work.* Strategic Management Journal. Accessed through the article appendix and companion materials at `https://sms.onlinelibrary.wiley.com/doi/full/10.1002/smj.3286`.

AIOE Data. *AIOE Data Repository.* GitHub repository. `https://github.com/AIOE-Data`.

Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, and J. Robert Warren. *Integrated Public Use Microdata Series, Current Population Survey: Version 12.0 [dataset].* Minneapolis, MN: IPUMS, 2025. The project uses CPS Annual Social and Economic Supplement (ASEC) files covering survey years 2016-2025.

U.S. Census Bureau. *2022 Census Industry Code List with Crosswalk.* Internal project file: `2022-Census-Industry-Code-List-with-Crosswalk.xlsx`.

U.S. Census Bureau. *2017 Industry Code List with Crosswalk.* Internal project file: `2017-industry-code-list-with-crosswalk.xlsx`.

U.S. Census Bureau. *2018 Occupation Code List and Crosswalk.* Internal project file: `2018-occupation-code-list-and-crosswalk.xlsx`.

For reproducibility within this repository, the main project files are `data_merging.ipynb`, `models/causal_analysis.ipynb`, `reports/data_science_memo.ipynb`, and `notebook_helpers.py`.